

FLARE Airfare Price Index

Methodology & Technical Reference

Complete explanation of how the FLARE Airfare Price Index is computed — from data collection through economic analysis. Every number displayed in the system is traceable to this methodology.

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1. Data Sources

OTA Booking Portals (via scrape.do)

Fares are collected from MakeMyTrip, Cleartrip, Ixigo, Yatra, Goibibo, and EaseMyTrip using the scrape.do JavaScript-rendering API. Each observation records the source portal, timestamp, and raw HTML selector path for provenance.

Government Airline — Air India

Air India is a Government of India (GoI) owned airline. Fares from airindia.com are tagged `source_type='govt_airline_scrape'` to distinguish them from private OTA data. This is significant for CPI analysis as it represents direct government pricing.

Direct Airline Portals

IndiGo direct bookings are collected from goindigo.in via scrape.do. These are tagged `source_type='airline_scrape'` and have no OTA convenience charge markup.

Amadeus GDS

The Amadeus Global Distribution System provides industry-standard fare data when API credentials are configured. Labelled `source_type='gds'`.

Google Flights (Apify)

Google Flights data via the Apify actor provides meta-search aggregated fares. Requires `APIFY_API_TOKEN`.

Demo Data

When live sources are unavailable, a DemoCollector generates statistically realistic synthetic fares. Always labelled `data_status='DEMO'` — never mixed with live data.

2. Collection Process

Trigger

Collection is triggered on every user search. All sources run in parallel via `asyncio.gather()`. Failures in any individual source do not affect others.

scrape.do Configuration

API: `api.scrape.do` · JS rendering enabled · `geoCode=IN` (India) · 5–6 second wait for React hydration · Retry on rate limits. Token: `0f737bcdd4d04085854df2da49a669e8a87e6d23cec`

Provenance Fields

Every observation records: `source` (e.g., 'makemytrip'), `source_type`, `observation_timestamp`, `data_status` (LIVE/DEMO/ERROR), `quality_score`.

3. Data Cleaning

Validation

Fares must be between ₹800 and ₹1,50,000. Travel date must be valid. Route must match known airport IATA codes. Invalid records are rejected and logged.

Deduplication

Observations with identical route, date, airline, and fare (rounded to nearest rupee) collected within the same 1-hour window are deduplicated. The original observation is kept; duplicates are discarded.

Outlier Detection

Z-score outlier detection applied per batch. Observations with $|Z| > 3.0$ are flagged (outlier_flag=True) but NOT deleted. Reason stored in outlier_reason. Quality score reduced.

4. Fare Normalization

Raw vs Normalized

Both raw_fare and normalized_fare are stored. Raw observations are never destroyed.

Convenience Charge Adjustment

OTA portals add convenience charges (₹150–₹350 typically). Known per-OTA estimates are subtracted from total_fare to derive the normalized fare. Direct airline sources (Air India, IndiGo) have zero convenience charge.

Stop Equivalency

Stop adjustments are recorded for reference but not applied to the index calculation — the index captures actual market prices including stop variations.

5. Route Selection

12-Route Basket

The index is calculated over a basket of 12 major domestic routes: DEL-BOM, BOM-DEL, DEL-BLR, BOM-BLR, DEL-HYD, BOM-MAA, DEL-MAA, BLR-HYD, DEL-CCU, BOM-CCU, DEL-COK, BOM-GOI.

Extensibility

The RouteBasket table is configurable. Additional routes can be added via the database with their respective weights.

6. Weighting

Weight Source

Route weights are based on approximate passenger traffic volumes from DGCA Annual Reports. The DEL-BOM corridor receives the highest weight (18%) as India's busiest route.

Weight Storage

Stored in route_basket table with fields: weight, weight_source, weight_methodology, effective_from, active.

Normalisation

Weights are normalised to sum to 1.0 across all active routes.

7. Index Calculation

Method: Laspeyres Price Index

$$\text{Index} = \frac{\sum (\text{current_route_avg} / \text{base_route_avg} \times 100) \times \text{route_weight}}{\sum \text{route_weight}}$$

Advance Purchase Window

Only fares for travel T+7 to T+30 days ahead are included in index calculation, standardising for advance-purchase effects.

Economy Direct/1-stop Only

Only economy class, maximum 1-stop fares are included to ensure comparability.

8. Base Period

Default Base Period

January 2025 = 100. This can be reconfigured in the system.

Base Fare

The average fare for each route during the base period is stored and used as the denominator in the Laspeyres formula.

9. Inflation Calculation

Year-on-Year (YoY)

$$\text{YoY Inflation} = \frac{(\text{Current Period Index} - \text{Same Month Prior Year Index})}{\text{Same Month Prior Year Index}} \times 100\%$$

Month-on-Month (MoM)

$$\text{MoM Inflation} = \frac{(\text{Current Month Index} - \text{Previous Month Index})}{\text{Previous Month Index}} \times 100\%$$

Minimum Data Requirement

YoY requires 13+ months of index data. MoM requires 2+ consecutive months. Values are not displayed until sufficient data exists.

10. CPI Comparison

Official CPI Source

Official CPI data from MoSPI/NSO (Ministry of Statistics & Programme Implementation, Government of India). Combined CPI, Rural+Urban, Base 2012=100.

Clear Labelling

The Computed Airfare Price Index and Official Government CPI are always displayed as separate series. They are never merged or presented as the same metric.

Inflation Gap

Gap = Airfare YoY% – Official CPI YoY%. A positive gap means airfares are inflating faster than general prices.

11. Contribution Methodology

Mode A — Official Weight Available

Estimated Contribution = Airfare Inflation (YoY%) × Official CPI Transport Weight (8.59%). Result expressed in percentage points (pp).

Mode B — No Official Weight

If no appropriate official weight is available, the system shows: 'Calculation unavailable — required official data/weight not available.' No contribution figure is fabricated.

Disclaimer

The estimated contribution is an analytical estimate based on the documented methodology. It is NOT an official Government of India CPI contribution figure. The NSO CPI basket does not publish a separate national weight for domestic air transport specifically.

12. Limitations

OTA Website Structure Changes

OTA websites frequently update their HTML structure. If scraping fails, the system falls back to DemoCollector and logs the failure — it does not crash.

scrape.do Rate Limits

The scrape.do API has rate limits. High-frequency collection across all sources may result in some requests being rate-limited.

CPI Weight Granularity

The NSO CPI basket does not provide a separate national weight for domestic air transport. The Transport & Communication weight (8.59%) is the most applicable official category available.

Base Period Sensitivity

The index value is sensitive to the choice of base period. A high-fare base period will produce a lower index for the same current fares.

Advance Purchase Standardisation

The T+7–T+30 window standardises for advance purchase but excludes last-minute and long-advance fares from the index.

Important Disclaimer: The FLARE system's Computed Airfare Price Index is an experimental, analytical tool produced from collected fare data. It is NOT an official Government of India statistic. Official CPI data is sourced from MoSPI/NSO and is always clearly labelled as such. The system maintains strict separation between live data, historical data, official government data, and synthetic demo data.